ChatGPT

The two–drug dosing problem is a simple linear program: minimize $x_1 + x_2$ subject to $x_1 + a_2 x_2 = 1$ with $x_i \geq 0$. In any LP, the optimum lies at a vertex of the feasible polytope [1]. Here one endpoint of the line $x_1 + a_2 x_2 = 1$ is $x_1 = 0, x_2 = 1/a_2$, the other is $x_2 = 0, x_1 = 1$. Which is best depends on a_2. From the code one sees

$$a_2 = 2f + \tfrac{1}{2}, \quad f = 72^{-10^{24}} + \sum_{k=3}^{\infty} \Big((1/2)^k - 72^{-10^{24} k} \Big).$$

Since $\sum_{k=3}^{\infty} (1/2)^k = 1/4$ [2] and the huge 72^{-1024} terms only add an extremely small positive amount, $f > 0.25$ so $a_2 > 1$. Thus the faster drug (rate a_2) should be used alone: the minimiser is $(x_1^*, x_2^*) = (0,\ 1/a_2)$.

Now compare $(0.1, 0.9)$ to $(0, 1/a_2)$. The ∞–norm distance is

$$\|(0.1, 0.9) - (0, 1/a_2)\|_\infty = \max \big(|0.1 - 0|,\ |0.9 - 1/a_2| \big).$$

By the ∞–norm definition, this is $\max(0.1,\ |0.9 - 1/a_2|)$ [3]. Since a_2 is only slightly above 1, $1/a_2 \approx 1$, so $|0.9 - 1/a_2| \approx 0.1$. In fact one checks $a_2 < 4/3$ (trivially, as $a_2 \approx 1$), so $|0.9 - 1/a_2| \leq 0.15$. Hence the distance is $\max(0.1, \approx 0.1) = 0.1 \leq 0.15$. In short, $(0.1, 0.9)$ is about 0.1 (in ℓ_∞) from the true minimiser, well within 0.15.

Answer: Yes. The optimal point is $(0, 1/a_2)$, and one finds

$$\|(0.1, 0.9) - (0, 1/a_2)\|_\infty = \max(0.1,\ |0.9 - 1/a_2|) \approx 0.1 \leq 0.15,$$

since a_2 (from the code) is just above 1 [2] [1] [3].

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[1] Linear programming - Wikipedia
https://en.wikipedia.org/wiki/Linear_programming

[2] Geometric series - Wikipedia
https://en.wikipedia.org/wiki/Geometric_series

[3] Norm (mathematics) - Wikipedia
https://en.wikipedia.org/wiki/Norm_(mathematics)